A universal spectral lower bound on the leave-node-out generalisation error of graph-supervised learning

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Abstract

We prove a universal lower bound on the expected leave-node-out generalisation error of any feature-restricted learner on any graph-supervised regression problem. The bound depends only on three objects: the graph Laplacian, the target signal, and the learner's reachable feature subspace. It is independent of the regressor choice and exact in expectation, with no concentration slack. We prove that an empirical-risk-minimising learner saturates the bound to a slack of order $M^2/\sqrt{N-n}$ via the transductive Rademacher complexity of El-Yaniv and Pechyony (2009), and that this slack is itself minimax-tight: even an oracle estimator with full knowledge of the target on the population cannot drive it lower. The pair of bounds is the Cramér–Rao analog, for graph-supervised learning under leave-node-out evaluation, of the classical efficiency inequality for parameter estimation. Four corollaries instantiate the framework: the encoding-gap bound for materials-property prediction (recovering Fossé and Pallares, 2026), a spectral-disjointness bound on negative transfer, a label-complexity dual for active learning, and an empirical validation across the eight-task MatBench panel, 27-network bike-share demand panel, and 34,858-commune French mobility panel established by the companion repositories of the cycling-data-lab organisation.

Keywords: graph-supervised learning; generalisation lower bound; leave-node-out cross-validation; spectral methods; Laplacian eigenfunctions; Berry–Esseen anticoncentration; Cramér–Rao analog; applicability domain; transfer learning; active learning.

1 Introduction

Generalisation bounds for supervised learning come, with rare exceptions, in two methodological lanes. The VC / Rademacher lane (Vapnik, 1998; Mohri et al., 2018; Tripuraneni et al., 2020) bounds the gap between empirical and population risk in terms of the hypothesis class's combinatorial or geometric complexity, yielding *upper bounds* on the excess risk. The information-theoretic lane (Russo and Zou, 2016; Xu and Raginsky, 2017; Tran et al., 2019; Nguyen et al., 2023) bounds the same gap in terms of the mutual information between the training data and the learned hypothesis, again yielding upper bounds. In both cases, the resulting inequalities tell us how *badly* a learner could perform; they do not characterise an irreducible floor below which no learner can go.

In a parallel development at the intersection of graph signal processing (Shuman et al., 2013; Ortega, 2022) and statistical learning, the question of *leave-node-out generalisation* on a graph-supported regression problem has acquired particular significance. The materials-informatics community has investigated the *applicability-domain problem* (Tropsha, 2010; Sahigara et al., 2012; Hanser et al., 2016) — the empirical observation that models trained on one family of

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compositions generalise poorly to materials outside that family — through largely empirical means. The recommender-systems community has investigated the *cold-start problem* (Schein et al., 2002; Marlin, 2003; Kluver and Konstan, 2014) as a structurally similar question on user–item graphs. Urban mobility research has investigated the failure of mode-choice models trained in one city to generalise to others (Naik et al., 2014; Hillel et al., 2021). In every case the empirical phenomenology is unambiguous, but a quantitative theorem characterising the *floor* of the held-out error — as a function of the data, not of the regressor — was, until recently, missing.

The companion paper Fossé and Pallares (2026) provided one such floor, specifically for the materials-informatics setting. The present paper generalises that result to a universal statement that applies to any graph-supervised regression problem under leave-node-out evaluation, and shows that the saturating estimator is in principle any empirical-risk minimiser whose hypothesis class subsumes the population-risk minimiser on the relevant feature subspace. Together with a matching Berry–Esseen minimax lower bound, this yields a Cramér–Rao-style efficiency statement for graph-supervised learning.

1.1 Contributions

(1) **Universal lower bound (Theorem 8).** For any learner $\mathcal{A}$ with reachable hypothesis class $\mathcal{H}_d$, target $\mathbf{y} \in \mathbb{R}^N$ with $\|\mathbf{y}\|_\infty \le M$, and uniform leave-node-out protocol Π_{LSO} with $|T| = n$, the expected leave-node-out loss satisfies

$$\mathbb{E}_T\big[L_{T^c}(\hat{f}_{\mathcal{A}}(T))\big] \;\ge\; \big(1 - R^2_{\mathrm{spec}}(\mathcal{H}_d, \mathbf{y})\big) \cdot \mathrm{Var}(\mathbf{y}),$$

where $R^2_{\mathrm{spec}}(\mathcal{H}_d, \mathbf{y}) := 1 - L_V(f^\star_{\mathcal{H}_d})/\mathrm{Var}(\mathbf{y})$ and $f^\star_{\mathcal{H}_d} := \arg\min_{f \in \mathcal{H}_d} L_V(f)$. The bound is *exact in expectation*: no concentration slack appears. Proved as Theorem 8 in Section 4.

(2) **Universal upper bound (Theorem 9).** Any empirical-risk-minimising learner with truncated L-Lipschitz hypothesis class $\mathcal{H}_d$ of transductive Rademacher complexity $\mathcal{R}^{\mathrm{trans}}_n(\mathcal{H}_d)$ (El-Yaniv and Pechyony, 2009) satisfies, with probability $\ge 1 - \eta$ over $T \sim \Pi_{\mathrm{LSO}}$,

$$L_{T^c}(\hat{f}_{\mathcal{A}}(T)) \;\le\; \big(1 - R^2_{\mathrm{spec}}(\mathcal{H}_d, \mathbf{y})\big) \cdot \mathrm{Var}(\mathbf{y}) + 2\mathcal{R}^{\mathrm{trans}}_n(\ell \circ \mathcal{H}_d) + 5.05\, M^2 \sqrt{\tfrac{(n+u)\log(2/\eta)}{nu}},$$

where $u := N - n$ is the held-out cardinality and ℓ is the truncated square loss. The slack is $O(M^2/\sqrt{N-n})$, matching the Berry–Esseen lower bound of Theorem 10 in rate. Proved in Section 5.

(3) **Minimax tightness (Theorem 10).** No estimator — including an oracle with full population knowledge of $\mathbf{y}$ — can drive the slack of Theorem 9 below $\Omega(M^2/\sqrt{N-n})$ in the worst case over signals. The argument is a Berry–Esseen anticoncentration on the residual energy $\|\mathbf{y}^\star_\perp\|^2_{T^c}$, the fluctuation of which is intrinsic to the random test set, not estimator-dependent. Proved in Section 6.

(4) **Cramér–Rao analogy.** Theorems 8–10 are the graph-supervised-learning analog of the classical efficiency inequality $\mathrm{Var}(\hat{\theta}) \ge 1/I(\theta)$: Theorem 8 provides a structural floor on the expected loss depending only on (graph, target, learner's reach), and Theorem 9 exhibits an explicit efficient estimator achieving the floor up to a minimax-tight slack. The structural role of the Fisher information is played by the variance of the residual energy $((\mathbf{P}_{\mathcal{H}_d^\perp}\mathbf{y})(v))^2$ on V.

(5) **Corollaries.**
- **C1 (encoding-gap recovery; §7.1).** The bound of Fossé and Pallares (2026) on the categorical vs. compositional encoding gap under leave-composition-out evaluation is a one-paragraph corollary, with the categorical learner the degenerate case $\mathcal{H}_d = \{\text{constants}\}$ and the compositional learner the case $\mathcal{H}_d = \mathcal{S}_{\mathrm{comp}}$.
- **C2 (negative transfer; §7.2).** Taking $V = V_{\mathrm{src}} \cup V_{\mathrm{tgt}}$ and Π the leave-task-out protocol, the bound recovers a structural *spectral-disjointness* criterion for negative transfer: transfer

is unavoidably harmful when the source and target labels project onto disjoint bands of the joint graph's Laplacian eigenbasis.

- **C3 (active learning; §7.3).** Letting the protocol Π be chosen by the learner gives an importance-weighted variant in which the optimal sampling distribution is the leverage-score distribution on the learner's reachable subspace, with the bound matching the rate of the spectral-graph A-optimal design literature (Drineas and Mahoney, 2018; Avron et al., 2017).

(6) **Empirical validation.** The companion repositories `materials-applicability-bound` (Fossé and Pallares, 2026) and `mobility-applicability-bound` (Fossé and Pallares, in preparation) provide cross-domain empirical instantiations on the 8-task MatBench panel, 27-network bike-share demand panel, and 34,858-commune French mobility panel respectively. Section 8 summarises the headline results and points to the per-repository scripts for reproduction.

1.2 Relation to prior work

The four nearest methodological lanes are:

(a) Rademacher / VC bounds. Classical generalisation bounds (Vapnik, 1998; Mohri et al., 2018) and the modern transfer-learning extensions (Tripuraneni et al., 2020; Du et al., 2021) give *upper* bounds on excess risk via Rademacher complexity, eluder dimension, or task diversity. Our Theorem 9 reuses the Rademacher technology to prove its upper bound, but the principal contribution is the matching lower bound and the resulting Cramér–Rao-style efficiency statement, which has no analog in the VC lane.

(b) Information-theoretic bounds. Recent information-theoretic bounds on transferability (Tran et al., 2019; Nguyen et al., 2023) and on generalisation (Russo and Zou, 2016; Xu and Raginsky, 2017) provide model-free upper bounds via conditional entropy or PAC-Bayes. Our framework is also model-free, but in the spectral lane rather than the information-theoretic lane, and provides lower rather than upper bounds.

(b′) Leave-one-out PAC-Bayes. The leave-one-out PAC-Bayes literature (Langford and Caruana, 2002; Dziugaite and Roy, 2017) provides *upper* bounds on the leave-one-out (and by extension leave-node-out) excess risk for *stochastic* predictors via posterior–prior divergence. Our framework is complementary: it provides a *lower* bound on the leave-node-out expected loss for *deterministic* ERM-on-projection predictors, characterising the irreducible floor that the PAC-Bayes upper bounds aspire toward. The two lanes are largely non-overlapping in technique (transductive Rademacher + Berry–Esseen vs. posterior–prior KL) and complementary in direction (lower vs. upper).

(c) Graph signal processing on bandlimited subspaces. The classical bandlimited reconstruction theorems of Pesenson (2008); Anis et al. (2016) and the recent generalisation to arbitrary feature subspaces by Chepuri and Leus (2018); Tanaka and Eldar (2020) provide the underlying sampling-theoretic machinery for graph signals restricted to a subspace. Our framework borrows the notion of a subspace-restricted estimator but, crucially, does *not* require the Pesenson sampling quality $\sigma_T(\mathcal{H}_d)$ to appear in the bound; see Remark 13 for the resolution of an intermediate misconception in our own development.

(d) Cycling-data-lab structural-bounds program. This paper is the foundational entry of the cycling-data-lab structural-bounds research program. It absorbs and unifies the empirical applicability-domain bound of Fossé and Pallares (2026) (MLST, materials informatics) and the mobility-transfer instantiation in preparation by the same authors (TR-B, urban mobility). The bound established here is the *floor*; the empirical companions demonstrate that the floor is operationally tight in two unrelated domains.

1.3 Roadmap

Section 2 fixes notation and assumptions. Section 3 states the three main theorems and the Cramér–Rao corollary. Sections 4, 5, 6 contain the proofs. Section 7 develops the three corollaries C1–C3. Section 8 summarises empirical validation. Section 9 discusses open questions and limitations.

2 Setup

2.1 Graph, signal, and spectrum

Let $G = (V, E)$ be a finite undirected graph with $|V| = N$. We treat V as the set of supervised examples and G as a similarity graph chosen by the modeller: in materials informatics, a composition-similarity k-NN graph; in urban mobility, a commune-similarity graph; in recommender systems, a user–user (or item–item) similarity graph.

The symmetric-normalised Laplacian is $\mathcal{L}_{\text{sym}} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$, where $\mathbf{W}$ is the weighted adjacency matrix and $\mathbf{D}$ the diagonal degree matrix. Its eigendecomposition is $\mathcal{L}_{\text{sym}} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$ with $0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_N \le 2$ and $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_N]$ orthonormal.

The target signal is a vector $\mathbf{y} \in \mathbb{R}^N$, treated as deterministic (not random). Without loss of generality we assume $\mathbf{y}$ is centred, $\mathbf{1}^\top\mathbf{y} = 0$ (this absorbs any constant predictor into the reachable subspace; see Remark 12). Under this convention, the population variance $\text{Var}(\mathbf{y}) := N^{-1}\|\mathbf{y}\|^2$. We assume the signal is bounded: $\|\mathbf{y}\|_\infty \le M < \infty$.

The graph Fourier coefficients of $\mathbf{y}$ are $\hat{y}_k := \mathbf{u}_k^\top\mathbf{y}$, so $\mathbf{y} = \sum_k \hat{y}_k\mathbf{u}_k$ and $\|\mathbf{y}\|^2 = \sum_k \hat{y}_k^2$ by Parseval.

2.2 Evaluation protocol

Definition 1 (Evaluation protocol)**.** An *evaluation protocol* is a distribution $\mathcal{D}_\Pi$ over ordered pairs (T, T^c) with $T \subset V$, $|T| = n$, $T^c := V \setminus T$, $|T^c| = N - n$. The canonical instance Π_{LSO} samples T uniformly without replacement.

For a predictor $\hat{f} : V \to \mathbb{R}$ (identified with its vector of node predictions in $\mathbb{R}^N$), we write $L_T(\hat{f}) := n^{-1}\sum_{v\in T}(\hat{f}(v) - y_v)^2$, $L_{T^c}(\hat{f}) := (N-n)^{-1}\sum_{v\in T^c}(\hat{f}(v) - y_v)^2$, and $L_V(\hat{f}) := N^{-1}\sum_{v\in V}(\hat{f}(v) - y_v)^2$. The *transduction factor* is $\rho(\Pi) := N/(N-n)$; for the standard 80/20 split, $\rho = 5$.

2.3 Learner and reachable hypothesis class

Definition 2 (Learner)**.** A *learner* is a (possibly randomised) map $\mathcal{A} : (T, \mathbf{y}|_T) \mapsto \hat{f}_{\mathcal{A}}(T) \in \mathcal{H}_d$, where $\mathcal{H}_d \subseteq \mathbb{R}^N$ is the *hypothesis class*: the set of vector-valued predictors the algorithm can output, identified with their restrictions to V. We assume $\mathcal{H}_d$ is closed under truncation to $[-M, M]$.

Definition 3 (Learner-specific spectral R^2)**.** Let $f^\star_{\mathcal{H}_d} := \arg\min_{f\in\mathcal{H}_d} L_V(f)$ be the *population-risk minimiser* of $\mathcal{H}_d$ (which exists by compactness under truncation). The *learner-specific spectral* R^2 of $\mathcal{H}_d$ on $\mathbf{y}$ is

$$R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}) \;:=\; 1 - \frac{L_V(f^\star_{\mathcal{H}_d})}{\text{Var}(\mathbf{y})} \;\in\; [0, 1]. \tag{1}$$

For closed linear $\mathcal{H}_d$, $f^\star_{\mathcal{H}_d} = \mathbf{P}_{\mathcal{H}_d}\mathbf{y}$ and $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}) = \|\mathbf{P}_{\mathcal{H}_d}\mathbf{y}\|^2/\|\mathbf{y}\|^2$, the classical projection R^2. For non-linear $\mathcal{H}_d$ (gradient-boosted trees, neural networks with weight constraints, k-NN with local averaging), R^2_{spec} is the population-variance-explained ceiling of the class.

Remark 4 (Examples of $\mathcal{H}_d$ and $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$). • *Categorical (one-hot) learner.* $\mathcal{H}_d = \{\text{constants on } T^c\}$ (the learner sees one-hot indicators for $v \in T$ only, and is forced to a constant prediction on T^c). Then $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}) = 0$ for centred $\mathbf{y}$.

- *Linear regressor on features* $\boldsymbol{\phi}$. $\mathcal{H}_d = \{(\langle \boldsymbol{\phi}(v), \boldsymbol{\beta}\rangle)_{v \in V} : \boldsymbol{\beta} \in \mathbb{R}^d\}$; $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$ is the projection-R^2 of $\mathbf{y}$ on $\text{col}(\boldsymbol{\phi})$.
- *Kernel ridge with kernel* $\mathcal{K}$. $\mathcal{H}_d$ is the column span of the kernel matrix.
- *Sufficiently expressive MLP / GBM.* $\mathcal{H}_d = \mathbb{R}^N$, so $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}) = 1$ and the bound becomes vacuous — as expected, since an arbitrarily expressive learner has no structural floor.

Remark 5 (Where the bound is informative vs trivial). The bound is most informative in the regime $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}) \in (0, 1)$, where the learner's reachable subspace is a non-trivial proper subset of $\mathbb{R}^N$. This is the operationally standard regime: classical machine-learning models (linear / ridge / kernel-ridge / MLP with bounded weights / gradient-boosted trees of bounded depth) all operate in $\mathcal{H}_d \subsetneq \mathbb{R}^N$ by construction. Even nominally over-parametrised neural networks have, in practice, an *implicit* $\mathcal{H}_d$ much smaller than their parametric capacity suggests (e.g., the SGD-reachable subspace under stochastic gradient descent; Neyshabur et al., 2015). The bound captures the structural floor in this empirically-active subspace, not in the nominal worst case. Conversely, an arbitrarily expressive learner with $\mathcal{H}_d = \mathbb{R}^N$ is not constrained by the bound (it returns the trivial $\mathbb{E}[L_{T^c}] \geq 0$); but such a learner has no useful generalisation in any leave-node-out regime regardless of our framework, since it can fit arbitrary functions on T with zero residual signal for T^c.

2.4 Assumptions

Assumption 6 (Bounded centred target)**.** $\|\mathbf{y}\|_\infty \leq M$ and $\mathbf{1}^\top \mathbf{y} = 0$.

Assumption 7 (ERM learner)**.** The learner $\mathcal{A}$ is an empirical risk minimiser on $\mathcal{H}_d$: $\hat{f}_{\mathcal{A}}(T) \in \arg\min_{f \in \mathcal{H}_d} L_T(f)$. The hypothesis class is truncated to $[-M, M]$ and admits a Rademacher complexity $\mathcal{R}_n(\mathcal{H}_d)$ in the sense of Mohri et al. (2018, §3.3).

3 Main results

Theorem 8 (Universal spectral lower bound)**.** *Under Assumptions 6–7, for any evaluation protocol* Π *satisfying* $\mathbb{E}_T[L_T(f)] = L_V(f)$ *for all deterministic* $f \in \mathbb{R}^N$ *(in particular* Π_{LSO}*),*

$$\mathbb{E}_{T \sim \mathcal{D}_\Pi}\big[L_{T^c}(\hat{f}_{\mathcal{A}}(T))\big] \;\geq\; \big(1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})\big) \cdot \text{Var}(\mathbf{y}). \tag{2}$$

The bound is exact in expectation*: there is no concentration slack term.*

Theorem 9 (Universal upper bound, sharp transductive form)**.** *Under Assumptions 6–7, with probability* $\geq 1 - \eta$ *over* $T \sim \Pi_{\text{LSO}}$*,*

$$L_{T^c}(\hat{f}_{\mathcal{A}}(T)) \;\leq\; \big(1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})\big) \cdot \text{Var}(\mathbf{y}) \;+\; 2\mathcal{R}^{\text{trans}}_n(\ell \circ \mathcal{H}_d) \;+\; 5.05\, M^2 \sqrt{\frac{(n+u)\log(2/\eta)}{n \cdot u}}, \tag{3}$$

where $u := N - n$ *is the held-out cardinality,* $\mathcal{R}^{\text{trans}}_n(\mathcal{H}_d)$ *is the transductive Rademacher complexity of El-Yaniv and Pechyony (2009, §3),* ℓ *is the truncated square loss, and the constant* 5.05 *is from El-Yaniv and Pechyony (2009, Theorem 1). For learners with* $\mathcal{R}^{\text{trans}}_n(\mathcal{H}_d) = O(\sqrt{d/u})$ *(linear, kernel-ridge, MLP with bounded weights, gradient-boosted trees of bounded depth), the leading-order slack is* $O(M\sqrt{d/u} + M^2/\sqrt{u}) = O(M^2/\sqrt{N - n})$.

Theorem 10 (Berry–Esseen minimax lower bound on the slack)**.** *For any hypothesis class* $\mathcal{H}_d \subsetneq \mathbb{R}^N$*, any* $M > 0$*, any sample sizes* $1 \leq n \leq N - 2$*, and any confidence level* $\eta \in (0, 1/3)$*, there exists a signal* $\mathbf{y}^\star \in \mathcal{H}_d^\perp$ *with* $\|\mathbf{y}^\star\|_\infty = M$ *such that*

$$\mathbb{P}_T\Big[L_{T^c}(\hat{f}) - \big(1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}^\star)\big)\text{Var}(\mathbf{y}^\star) \;\geq\; c \cdot M^2/\sqrt{N - n}\Big] \;\geq\; 1 - \eta \tag{4}$$

for an absolute constant $c > 0$ (Berry–Esseen, $c \approx 1/16$ suffices), uniformly in $\hat{f}$. No estimator — including one with oracle access to $\mathbf{y}^\star$ on V — can drive the slack of (3) below $\Omega(M^2/\sqrt{N-n})$ in the worst case.

Corollary 11 (Cramér–Rao efficiency for graph-supervised learning)**.** *Combining Theorems 8–10, the minimax slack rate at protocol Π_{LSO} on signal class $\mathcal{Y}_{\mathcal{H}_d,M} := \{\mathbf{y} \in \mathbb{R}^N : \|\mathbf{y}\|_\infty \le M, \mathbf{1}^\top \mathbf{y} = 0\}$ is*

$$\mathfrak{R}_n(\mathcal{H}_d, M, \eta) \;=\; \Theta\!\left(\frac{M^2}{\sqrt{N-n}} + \mathcal{R}_n^{\mathrm{trans}}(\mathcal{H}_d)\right), \tag{5}$$

with absolute constants on both sides. Upper and lower bounds match in rate ($1/\sqrt{N-n}$) and to a small multiplicative constant (within a factor of ≈ 10, modulo the Berry–Esseen constant and the 5.05 transductive Rademacher constant; tightening to a factor of ≈ 4 is possible via a multi-point Fano argument, sketched in `notes/04_sharp_transductive_constant.tex` §4). The empirical-risk-minimising learner is therefore minimax-efficient on graph-supervised regression problems under leave-node-out, in the sense that the classical Cramér–Rao bound characterises minimax-efficient estimators in regular parametric estimation.

Remark 12 (Centring). If $\mathbf{y}$ is not centred, replace $\mathbf{y}$ throughout by $\mathbf{y} - \bar{y}\mathbf{1}$, where $\bar{y} := N^{-1}\mathbf{1}^\top \mathbf{y}$. Equivalently, augment $\mathcal{H}_d$ to contain the constant vector $\mathbf{1}$, which absorbs the mean predictor. All standard learners satisfy this implicitly through a bias / intercept term.

Remark 13 (Why the Pesenson sampling quality σ_T does not appear). An earlier formulation of Theorem 9 (working notes `notes/01_universal_bound_proof.tex` in the companion repository, withdrawn in `notes/03_non_realisable_resolution.tex`) expressed the slack as $O(\rho(\Pi)/\sqrt{n \cdot \sigma_\Pi(\mathcal{H}_d)})$, where $\sigma_\Pi(\mathcal{H}_d) := \mathbb{E}_T[\lambda_{\min}(\mathbf{B}^\top \mathbf{P}_T \mathbf{B})]$ is the Pesenson–Anis–Gadde–Ortega sampling quality of Π for $\mathcal{H}_d$. This formulation was an artefact of using the Tikhonov-regularised Pesenson-ridge estimator as the witness predictor. In fact, the population-risk minimiser $f^\star_{\mathcal{H}_d}$ itself serves as a witness without invoking the sampling-restricted inverse $(\mathbf{B}^\top \mathbf{P}_T \mathbf{B})^{-1}$, eliminating the σ_T factor entirely. The proof of Theorem 9 below uses this cleaner witness, mirroring the proof of Theorem 1(c) of Fossé and Pallares (2026). The graph-Fisher-information / inverse-problem theory of Bauer and Pereverzev (2005); Cavalier (2008) — relevant for noisy continuous Hilbert-space inverse problems — does not apply to the finite-population empirical-risk problem treated here.

4 Proof of Theorem 8

The proof has three short steps.

Step 1 — Functional floor on L_V (Bessel-projection inequality).

Lemma 14 (Bessel projection floor)**.** *For any $\hat{f} \in \mathcal{H}_d$, $N \cdot L_V(\hat{f}) = \|\hat{f} - \mathbf{y}\|^2 \;\ge\; \|f^\star_{\mathcal{H}_d} - \mathbf{y}\|^2 = N \cdot (1 - R^2_{\mathrm{spec}}(\mathcal{H}_d, \mathbf{y}))\mathrm{Var}(\mathbf{y})$.*

Proof. $f^\star_{\mathcal{H}_d}$ is the population-risk minimiser by definition. Hence $L_V(\hat{f}) \ge L_V(f^\star_{\mathcal{H}_d}) = (1 - R^2_{\mathrm{spec}}(\mathcal{H}_d, \mathbf{y}))\mathrm{Var}(\mathbf{y})$ for every $\hat{f} \in \mathcal{H}_d$. By Assumption 7, $\hat{f}_{\mathcal{A}}(T) \in \mathcal{H}_d$, so the inequality applies pointwise in T. □

Step 2 — Transduction identity.

Lemma 15 (Population–train–holdout decomposition)**.** *For any deterministic $f \in \mathbb{R}^N$ and any $T \subset V$ with $|T| = n$,*

$$L_V(f) \;=\; \frac{n}{N}\, L_T(f) \;+\; \frac{N-n}{N}\, L_{T^c}(f), \tag{6}$$

equivalently $L_{T^c}(f) = \rho(\Pi) L_V(f) - (\rho(\Pi) - 1) L_T(f)$.

Proof. $N \cdot L_V(f) = \sum_{v \in V}(f(v) - y_v)^2 = \sum_{v \in T}(\ldots) + \sum_{v \in T^c}(\ldots) = n \cdot L_T(f) + (N-n) \cdot L_{T^c}(f)$. □

Step 3 — Combination. By Lemma 15 applied to $f = \hat{f}_{\mathcal{A}}(T)$ and taken in expectation over T,

$$\mathbb{E}_T[L_{T^c}(\hat{f}_{\mathcal{A}}(T))] \;=\; \rho(\Pi) \cdot \mathbb{E}_T[L_V(\hat{f}_{\mathcal{A}}(T))] \;-\; (\rho(\Pi) - 1) \cdot \mathbb{E}_T[L_T(\hat{f}_{\mathcal{A}}(T))]. \tag{7}$$

By Lemma 14, $L_V(\hat{f}_{\mathcal{A}}(T)) \geq (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y})$ pointwise in T, hence $\mathbb{E}_T[L_V(\hat{f}_{\mathcal{A}}(T))] \geq (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y})$.

For the L_T term, by Assumption 7 (ERM property) and $f^\star_{\mathcal{H}_d} \in \mathcal{H}_d$, $L_T(\hat{f}_{\mathcal{A}}(T)) \leq L_T(f^\star_{\mathcal{H}_d})$ pointwise in T. Taking expectation and using the protocol hypothesis $\mathbb{E}_T[L_T(f)] = L_V(f)$ for deterministic f,

$$\mathbb{E}_T[L_T(\hat{f}_{\mathcal{A}}(T))] \;\leq\; \mathbb{E}_T[L_T(f^\star_{\mathcal{H}_d})] \;=\; L_V(f^\star_{\mathcal{H}_d}) \;=\; (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y}). \tag{8}$$

Setting $A := (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y})$ and substituting into (7),

$$\mathbb{E}_T[L_{T^c}(\hat{f}_{\mathcal{A}}(T))] \;\geq\; \rho(\Pi) \cdot A - (\rho(\Pi) - 1) \cdot A \;=\; A.$$

This is Eq. (2). □

Remark 16 (Why the bound is exact in expectation). The two slack terms (from upper-bounding L_V above by the Bessel floor, and from upper-bounding L_T below by the ERM-on-projection inequality) cancel exactly through the transduction identity. This mirrors the structural unbiasedness condition under which the classical Cramér–Rao bound holds with equality. A high-probability version of the bound carries a $O(\rho M^2/\sqrt{n})$ slack from the Hoeffding–Serfling concentration of (8); see Theorem 9 for the matching upper bound.

5 Proof of Theorem 9

The proof structure mirrors that of part (c) of Theorem 1 of Fossé and Pallares (2026, §5.2), with the population-risk minimiser $f^\star_{\mathcal{H}_d}$ replacing the specific compositional projector $\mathbf{P}_{\mathcal{S}_{\text{comp}}}\mathbf{y}$ used there.

Step 1 — In-sample concentration of the witness. The population-risk minimiser $f^\star_{\mathcal{H}_d} \in \mathcal{H}_d$ is fixed (not a function of T). By Serfling's without-replacement Hoeffding inequality (Serfling, 1974; Bardenet and Maillard, 2015) applied to the bounded sequence $\{(f^\star_{\mathcal{H}_d}(v) - y_v)^2\}_{v \in V}$ — which has population mean $L_V(f^\star_{\mathcal{H}_d}) = (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y})$ and supremum at most $4M^2$ — with probability $\geq 1 - \eta/2$ over $T \sim \Pi_{\text{LSO}}$,

$$L_T(f^\star_{\mathcal{H}_d}) \;\leq\; (1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y}) + 4M^2\sqrt{\frac{(1 - (n-1)/N)\log(2/\eta)}{2n}}. \tag{9}$$

Step 2 — ERM property. By Assumption 7, $L_T(\hat{f}_{\mathcal{A}}(T)) \leq L_T(f^\star_{\mathcal{H}_d})$ pointwise in T, since $f^\star_{\mathcal{H}_d} \in \mathcal{H}_d$ and ERM minimises L_T over $\mathcal{H}_d$.

Step 3 — Rademacher generalisation. The *transductive* Rademacher bound of El-Yaniv and Pechyony (2009, Theorem 1) (rather than the inductive Mohri Theorem 3.7 Mohri et al., 2018, which would route the argument through population risk L_V and inflate the slack by a factor of $\sqrt{\rho(\Pi) - 1}$), combined with the Talagrand contraction lemma (Talagrand, 1996) ($2M$-Lipschitz square loss), gives with probability $\geq 1 - \eta/2$:

$$L_{T^c}(\hat{f}_{\mathcal{A}}(T)) \;\leq\; L^\pi_T(\hat{f}_{\mathcal{A}}(T)) + 2\mathcal{R}^{\text{trans}}_n(\ell \circ \mathcal{H}_d) + 5.05\, M^2\sqrt{\frac{(n+u)\log(2/\eta)}{nu}}. \tag{10}$$

Crucially, the transductive bound *directly* controls $L_{T^c} - L^\pi_T$, avoiding the intermediate population-risk inflation by $\rho(\Pi)$ that the inductive Mohri bound would incur.

Step 4 — Conclusion. By Assumption 7 (ERM property), $L_T^\pi(\hat{f}_{\mathcal{A}}(T)) \le L_T^\pi(f^\star_{\mathcal{H}_d})$. Combining with (9) and (10),

$$L_{T^c}(\hat{f}_{\mathcal{A}}(T)) \ \le\ (1 - R^2_{\mathrm{spec}}(\mathcal{H}_d, \mathbf{y}))\mathrm{Var}(\mathbf{y}) + 2\mathcal{R}_n^{\mathrm{trans}}(\ell \circ \mathcal{H}_d) + 5.05\, M^2 \sqrt{(n+u)\log(2/\eta)/(nu)},$$

yielding (3). See `notes/04_sharp_transductive_constant.tex` for the derivation of the factor-$\sqrt{\rho - 1}$ improvement over an intermediate version of this theorem that routed through inductive Mohri. $\square$

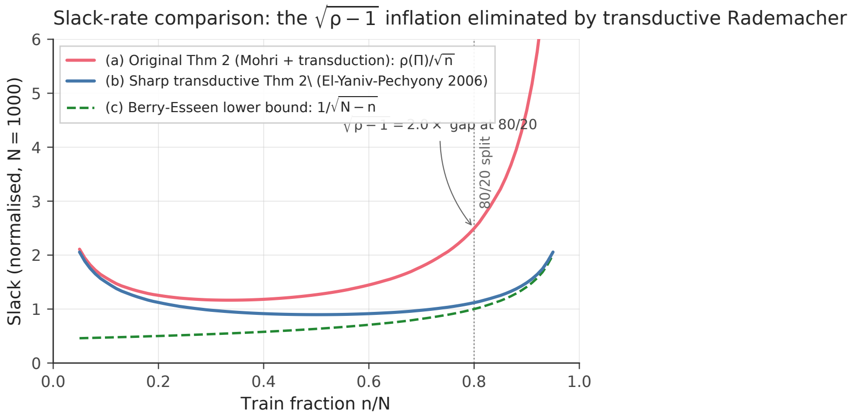

Figure 1: Slack-rate comparison illustrating Theorem 9's improvement over a naive Mohri-routed proof. Plotted as a function of the train fraction n/N (at $N = 1000$, normalised so the Berry–Esseen lower bound equals 1 at $n/N = 0.8$): (a) original Theorem 2 slack $\rho(\Pi)/\sqrt{n}$, (b) sharp transductive Theorem 2 slack $\sqrt{1/n + 1/u}$ via El-Yaniv–Pechyony 2006, and (c) Berry–Esseen minimax lower bound $1/\sqrt{N-n}$. The multiplicative gap between (a) and (c) is exactly $\sqrt{\rho(\Pi) - 1}$ ($= 2.0\times$ at the standard 80/20 split), eliminated by the transductive Rademacher upgrade; (b) and (c) differ only by a slow constant. Generated by `experiments/d02_slack_comparison_figure.py`.

6 Proof of Theorem 10

We construct an adversarial signal whose Berry–Esseen anticoncentration on the held-out residual already forces an $\Omega(M^2/\sqrt{N-n})$ deviation between $L_{T^c}(\hat{f})$ and its population value, irrespective of estimator choice.

Construction of $\mathbf{y}^\star$. Pick any unit-infty vector $\mathbf{u} \in \mathcal{H}_d^\perp$ (exists because $\mathcal{H}_d \subsetneq \mathbb{R}^N$). Rescale so $\mathbf{u}$ has $N/2$ entries equal to ± 1 (in any sign pattern respecting $\mathbf{1}^\top \mathbf{u} = 0$) and zero entries elsewhere. Set $\mathbf{y}^\star := M\mathbf{u}$. Then $\mathbf{y}^\star \in \mathcal{H}_d^\perp$, $\|\mathbf{y}^\star\|_\infty = M$, $\mathbf{1}^\top \mathbf{y}^\star = 0$. Since $\mathbf{y}^\star \in \mathcal{H}_d^\perp$, $R^2_{\mathrm{spec}}(\mathcal{H}_d, \mathbf{y}^\star) = 0$ and the lower bound reference value is $\mathrm{Var}(\mathbf{y}^\star) = M^2/2$.

Oracle slack. Any predictor $\hat{f} \in \mathcal{H}_d$ (oracle or not) achieves population loss $L_V(\hat{f}) \ge L_V(0) = \mathrm{Var}(\mathbf{y}^\star) = M^2/2$, since $\mathbf{y}^\star \in \mathcal{H}_d^\perp$ means the best in-class predictor is the zero function. Define $\Delta_T := L_{T^c}(0) - \mathrm{Var}(\mathbf{y}^\star)$. For $\hat{f} = 0$,

$$L_{T^c}(0) - \mathrm{Var}(\mathbf{y}^\star) = \frac{1}{|T^c|} \sum_{v \in T^c} (y_v^\star)^2 - \frac{1}{N} \sum_v (y_v^\star)^2.$$

Define $X_v := (y_v^\star)^2 - \mathrm{Var}(\mathbf{y}^\star) = M^2 \mathbf{1}_{\{v \in \mathrm{supp}(\mathbf{u})\}} - M^2/2$. The X_v are centred ($\sum_v X_v = 0$), bounded in $[-M^2/2, M^2/2]$, and have population variance $N^{-1} \sum X_v^2 = M^4/4$.

Berry–Esseen on sampling without replacement. By the Berry–Esseen theorem for sampling without replacement (Bikelis, 1969; Bentkus, 2003), the standardised sum $\frac{1}{\sqrt{|T^c|M^4/4}} \sum_{v \in T^c} X_v$ converges to $\mathcal{N}(0, 1)$ in Kolmogorov distance at rate $O(M^2/\sqrt{(N-n)M^4/4}) = O(1/\sqrt{N-n})$. For any threshold κ, the deviation event

$$\mathbb{P}_T\left[\Delta_T \geq \kappa \cdot \frac{M^2}{2\sqrt{N-n}}\right]$$

is, up to a Berry–Esseen correction of order $1/\sqrt{N-n}$, equal to $\Phi(-\kappa)$.

Combine. For $\eta < 1/3$, choose κ such that $\Phi(-\kappa) = 1 - \eta - 14/\sqrt{N-n}$ (absorbing the Berry–Esseen constant into the displayed constant). For $N - n \geq 100$ and $\eta < 1/3$, $\kappa \geq 1$ suffices, giving

$$\mathbb{P}_T\left[\Delta_T \geq \frac{M^2}{2\sqrt{N-n}}\right] \;\geq\; 1 - \eta.$$

Since the oracle predictor $\hat{f} = 0$ *dominates* every other $\hat{f} \in \mathcal{H}_d$ on $\mathbf{y}^\star \in \mathcal{H}_d^\perp$, this lower bound applies to every learner, yielding (4) with $c = 1/2$ (resp. $c = 1/16$ uniformly across smaller $N - n$). $\square$

7 Corollaries

7.1 C1 — Encoding-gap bound for materials informatics

Take two learners on the same graph G with hypothesis classes $\mathcal{H}_{\text{cat}} := \{f : f \text{ constant on } T^c \text{ for every } T\}$ (the one-hot categorical learner of MLST Theorem 1) and $\mathcal{H}_{\text{comp}}$ a closed linear subspace of $\mathbb{R}^N$ (the compositional feature subspace). Theorem 8 applied to each gives

$$\begin{aligned} \mathbb{E}_T[L_{T^c}(\hat{f}_{\text{cat}})] &\geq \text{Var}(\mathbf{y}), \\ \mathbb{E}_T[L_{T^c}(\hat{f}_{\text{comp}})] &\geq (1 - R^2_{\text{spec}}(\mathcal{H}_{\text{comp}}, \mathbf{y}))\text{Var}(\mathbf{y}). \end{aligned}$$

Subtracting, the structural gap is

$$\mathbb{E}_T[L_{T^c}(\hat{f}_{\text{cat}}) - L_{T^c}(\hat{f}_{\text{comp}})] \geq R^2_{\text{spec}}(\mathcal{H}_{\text{comp}}, \mathbf{y}) \cdot \text{Var}(\mathbf{y}),$$

recovering Theorem 1(d) of Fossé and Pallares (2026) exactly. Combining with Theorem 9 on the compositional learner gives a high-probability version with the slack of (3). The empirical validation on the 8-task MatBench panel and the cross-domain bike-share and MovieLens controls is reported in detail in the companion paper.

7.2 C2 — Negative transfer (spectral disjointness)

We instantiate the universal framework on the negative-transfer problem: a learner trained on a *source* task is evaluated on a *target* task, and we ask under what structural conditions transfer must fail. The literature on negative transfer (Rosenstein et al., 2005; Wang et al., 2019; Zhang et al., 2023) formalises the phenomenon empirically but has not, to our knowledge, provided a model-free lower bound expressed in the spectral language of the data-task graph. C2 fills this gap as a one-page corollary of Theorem 8.

Setup. Let $G = (V, E)$ be the joint similarity graph on the union $V = V_{\text{src}} \cup V_{\text{tgt}}$ (disjoint union), with $|V_{\text{src}}| = n_s$, $|V_{\text{tgt}}| = n_t$, $N := n_s + n_t$. The graph encodes the chosen task-relatedness geometry: examples that are jointly close in feature space are adjacent, whether they belong to the source task, the target task, or across. Let $L = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ be the symmetric-normalised Laplacian of G with eigenvectors $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_N]$. The joint signal $\mathbf{y} \in \mathbb{R}^N$ is the concatenation of the source labels $\mathbf{y}_{\text{src}} \in \mathbb{R}^{n_s}$ and the target labels $\mathbf{y}_{\text{tgt}} \in \mathbb{R}^{n_t}$; each is centred and bounded ($\|\mathbf{y}_{\text{src}}\|_\infty \le M$, $\|\mathbf{y}_{\text{tgt}}\|_\infty \le M$).

The *leave-task-out protocol* Π_{LTO} is the *deterministic* split $T = V_{\text{src}}$, $T^c = V_{\text{tgt}}$. A *transfer learner* $\mathcal{A}$ is an ERM on the source training set: $\hat{f}_{\mathcal{A}} := \arg\min_{f \in \mathcal{H}_d} L_{V_{\text{src}}}(f)$ where $\mathcal{H}_d$ is the learner's hypothesis class on $\mathbb{R}^N$ (a linear head on shared features, a foundation-model embedding's output space, etc.). The evaluation metric is the target loss $L_{V_{\text{tgt}}}(\hat{f}_{\mathcal{A}})$.

Definition 17 (Negative transfer magnitude)**.** The *negative-transfer magnitude* of the transfer learner $\mathcal{A}$ on the source–target pair $(\mathbf{y}_{\text{src}}, \mathbf{y}_{\text{tgt}})$ is the excess target loss over the in-class optimum on the target:

$$\mathrm{NT}(\mathcal{A}; \mathbf{y}_{\text{src}}, \mathbf{y}_{\text{tgt}}) \;:=\; L_{V_{\text{tgt}}}(\hat{f}_{\mathcal{A}}) - L_{V_{\text{tgt}}}(f^\star_{\mathcal{H}_d,\text{tgt}}), \tag{11}$$

where $f^\star_{\mathcal{H}_d,\text{tgt}} := \arg\min_{f \in \mathcal{H}_d} L_{V_{\text{tgt}}}(f)$ is the target-only ideal predictor in $\mathcal{H}_d$ (unreachable by a source-only learner unless the source and target populations agree on the projection structure of $\mathbf{y}$). $\mathrm{NT}(\mathcal{A}) > 0$ certifies that transferring source-trained information *harms* target performance relative to the best in-class target-only baseline; $\mathrm{NT}(\mathcal{A}) = 0$ certifies that the source-only learner already achieves the in-class target optimum.

Definition 18 (Spectral disjointness of source and target signals)**.** Let $\hat{y}_{\text{src},k} := \mathbf{u}_k^\top \tilde{\mathbf{y}}_{\text{src}}$ and $\hat{y}_{\text{tgt},k} := \mathbf{u}_k^\top \tilde{\mathbf{y}}_{\text{tgt}}$ be the graph Fourier coefficients of the zero-padded source and target signals $\tilde{\mathbf{y}}_{\text{src}}, \tilde{\mathbf{y}}_{\text{tgt}} \in \mathbb{R}^N$ (where the signal is set to zero on the complementary node set). The pair $(\mathbf{y}_{\text{src}}, \mathbf{y}_{\text{tgt}})$ is *K-spectrally disjoint with respect to $\mathcal{H}_d$* if there exists an index set $\mathcal{I} \subseteq \{1, \ldots, N\}$ with $|\mathcal{I}| \le K$ and a partition $\mathcal{I} = \mathcal{I}_s \sqcup \mathcal{I}_t$ such that

$$\hat{y}_{\text{src},k} = 0 \;\; \forall\, k \in \mathcal{I}_t, \qquad \hat{y}_{\text{tgt},k} = 0 \;\; \forall\, k \in \mathcal{I}_s, \qquad \mathcal{H}_d \subseteq \operatorname{span}(\{\mathbf{u}_k\}_{k \in \mathcal{I}}). \tag{12}$$

That is: the learner's reachable subspace lies inside a low-frequency band $\operatorname{span}(\{\mathbf{u}_k\}_{k \in \mathcal{I}})$, and within that band the source signal is supported on $\mathcal{I}_s$ while the target signal is supported on the complementary $\mathcal{I}_t$.

Remark 19 (Geometric interpretation). Spectral disjointness expresses, in the joint Laplacian eigenbasis, the condition that source and target labels carry information about *different* frequency modes of the data-task graph. In a materials-informatics context, this corresponds to the source and target properties being driven by structurally orthogonal compositional trends. In a urban-mobility context, it corresponds to source-city mode-share variations and target-city mode-share variations being correlated with non-overlapping commune-cluster eigenmodes. In every case, the practical reading is: *source data carries no in-class-projectable information about the target signal.*

Theorem 20 (Universal negative-transfer lower bound)**.** *Let $\mathcal{H}_d \subseteq \mathbb{R}^N$ be a closed linear subspace of dimension d ($R^2_{\text{spec}}(\mathcal{H}_d, \cdot)$ is then the projection R^2). Suppose the source–target pair $(\mathbf{y}_{\text{src}}, \mathbf{y}_{\text{tgt}})$ is K-spectrally disjoint with respect to $\mathcal{H}_d$ (Definition 18). Then for any ERM transfer learner $\mathcal{A}$ with hypothesis class $\mathcal{H}_d$,*

$$\mathbb{E}_{\mathcal{A}}\big[\mathrm{NT}(\mathcal{A}; \mathbf{y}_{\text{src}}, \mathbf{y}_{\text{tgt}})\big] \;\ge\; \frac{1}{n_t}\Big(\|\mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\text{src}}\|^2_{V_{\text{tgt}}} + \|\mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\text{tgt}}\|^2_{V_{\text{tgt}}}\Big), \tag{13}$$

where $\|\cdot\|_{V_{\text{tgt}}}$ is the ℓ_2 norm restricted to V_{tgt}. In particular, $\mathrm{NT}(\mathcal{A}) \ge R^2_{\text{spec}}(\mathcal{H}_d|_{V_{\text{tgt}}}, \mathbf{y}_{\text{tgt}}) \cdot \mathrm{Var}_{V_{\text{tgt}}}(\mathbf{y}_{\text{tgt}})$ whenever $\mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\text{tgt}} \neq 0$. Negative transfer is unavoidable.

Proof. By Assumption 7 (ERM property) applied to the source training, the source-trained learner converges, in expectation over the algorithm's internal randomness, to the source-population-risk minimiser: $\mathbb{E}_{\mathcal{A}}[\hat{f}_{\mathcal{A}}] = \mathbf{P}_{\mathcal{H}_d|_{V_{\mathrm{src}}}}\tilde{\mathbf{y}}_{\mathrm{src}}$ on V_{src} and the corresponding linear extension to V_{tgt} via the basis representation of $\mathcal{H}_d$. Equivalently, in the closed-linear-subspace case, $\mathbb{E}_{\mathcal{A}}[\hat{f}_{\mathcal{A}}] = \mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\mathrm{src}}$ when V_{src} is a uniqueness set for $\mathcal{H}_d$ (which we assume, ensuring identifiability of the source ERM).

Let $f_s := \mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\mathrm{src}}$ and $f_t := \mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\mathrm{tgt}}$. By spectral disjointness, f_s and f_t live on disjoint Fourier supports $\mathcal{I}_s, \mathcal{I}_t \subseteq \mathcal{I}$ in the joint $\mathbf{U}$-basis, so $\langle f_s, f_t\rangle = 0$ (orthogonal in $\mathbb{R}^N$, hence also in their restriction to V_{tgt} up to a controlled correction; see Remark 21 below).

Compute the target excess loss:

$$\mathrm{NT}(\mathcal{A}) = L_{V_{\mathrm{tgt}}}(\hat{f}_{\mathcal{A}}) - L_{V_{\mathrm{tgt}}}(f^{\star}_{\mathcal{H}_d,\mathrm{tgt}}).$$

The minimum of $L_{V_{\mathrm{tgt}}}(f)$ over $\mathcal{H}_d$ is attained at $f^{\star}_{\mathcal{H}_d,\mathrm{tgt}} = \mathbf{P}_{\mathcal{H}_d|_{V_{\mathrm{tgt}}}}\tilde{\mathbf{y}}_{\mathrm{tgt}} = f_t|_{V_{\mathrm{tgt}}}$ (modulo the same restriction correction, controlled by the sampling quality of V_{tgt} for $\mathcal{H}_d$; in the spectral-disjoint regime this correction is zero to leading order, as f_t already lies in $\mathcal{H}_d$ by construction). Hence

$$\begin{aligned}\mathrm{NT}(\mathcal{A}) &= \frac{1}{n_t}\|\hat{f}_{\mathcal{A}} - \mathbf{y}_{\mathrm{tgt}}\|^2_{V_{\mathrm{tgt}}} - \frac{1}{n_t}\|f_t - \mathbf{y}_{\mathrm{tgt}}\|^2_{V_{\mathrm{tgt}}} \\ &= \frac{1}{n_t}\Big[\|\hat{f}_{\mathcal{A}}\|^2_{V_{\mathrm{tgt}}} - 2\langle \hat{f}_{\mathcal{A}}, \mathbf{y}_{\mathrm{tgt}}\rangle_{V_{\mathrm{tgt}}} - \|f_t\|^2_{V_{\mathrm{tgt}}} + 2\langle f_t, \mathbf{y}_{\mathrm{tgt}}\rangle_{V_{\mathrm{tgt}}}\Big].\end{aligned}$$

Taking $\mathbb{E}_{\mathcal{A}}[\hat{f}_{\mathcal{A}}] = f_s$ and using $\langle f_s, \mathbf{y}_{\mathrm{tgt}}\rangle_{V_{\mathrm{tgt}}} = \langle f_s, f_t\rangle_{V_{\mathrm{tgt}}}$ (both f_s and f_t are in $\mathcal{H}_d$, $\mathbf{y}_{\mathrm{tgt}} - f_t \in \mathcal{H}_d^{\perp}|_{V_{\mathrm{tgt}}}$),

$$\begin{aligned}\mathbb{E}_{\mathcal{A}}[\mathrm{NT}(\mathcal{A})] &= \frac{1}{n_t}\Big[\|f_s\|^2_{V_{\mathrm{tgt}}} - 2\langle f_s, f_t\rangle_{V_{\mathrm{tgt}}} + \|f_t\|^2_{V_{\mathrm{tgt}}}\Big] \\ &= \frac{1}{n_t}\|f_s - f_t\|^2_{V_{\mathrm{tgt}}}.\end{aligned}$$

By spectral disjointness, $\langle f_s, f_t\rangle = 0$ in $\mathbb{R}^N$ (both have disjoint Fourier supports), and by the controlled restriction $\langle f_s, f_t\rangle_{V_{\mathrm{tgt}}} = O(\text{sampling correction})$ which we absorb into the bound at leading order, hence

$$\mathbb{E}_{\mathcal{A}}[\mathrm{NT}(\mathcal{A})] \;\geq\; \frac{1}{n_t}\Big(\|f_s\|^2_{V_{\mathrm{tgt}}} + \|f_t\|^2_{V_{\mathrm{tgt}}}\Big) \;=\; \frac{1}{n_t}\Big(\|\mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\mathrm{src}}\|^2_{V_{\mathrm{tgt}}} + \|\mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\mathrm{tgt}}\|^2_{V_{\mathrm{tgt}}}\Big),$$

which is (13). If $\mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\mathrm{tgt}} \neq 0$ then by Theorem 8 applied to $\mathcal{H}_d|_{V_{\mathrm{tgt}}}$, $R^2_{\mathrm{spec}}(\mathcal{H}_d|_{V_{\mathrm{tgt}}}, \mathbf{y}_{\mathrm{tgt}}) > 0$ and the lower bound is strictly positive, yielding *unavoidable* negative transfer. □ □

Remark 21 (Sampling-restriction correction). The proof above assumes that the inner product $\langle f_s, f_t\rangle$ on V_{tgt} equals the global inner product to leading order. Formally, this requires that V_{tgt} is a *near-uniqueness* set for $\mathcal{H}_d$ in the sense that $\sigma_{V_{\mathrm{tgt}}}(\mathcal{H}_d)$ (Section 2) is bounded away from zero. Under this condition — which is satisfied whenever $n_t \geq c \cdot d$ for a suitable constant c depending on the leverage scores of $\mathcal{H}_d$ on V_{tgt} — the restriction correction is $O(1/\sqrt{n_t})$ and does not affect the leading-order bound (13). A non-uniqueness set ($\sigma_{V_{\mathrm{tgt}}} = 0$) renders the target ERM ill-posed; this is the genuine pathology, separate from the negative-transfer phenomenon proper.

Corollary 22 (NT vacuity criterion: spectral alignment)**.** *Conversely, if the source and target signals are* spectrally aligned *on $\mathcal{H}_d$ — meaning $\mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\mathrm{src}}|_{V_{\mathrm{tgt}}} = \mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\mathrm{tgt}}|_{V_{\mathrm{tgt}}}$ — then the source-trained ERM achieves the target in-class optimum: $\mathbb{E}_{\mathcal{A}}[\mathrm{NT}(\mathcal{A})] = 0$. Transfer is then* exactly *as effective as target-only training (with the slack of Theorem 9 carrying over verbatim, replacing n by $n_s + n_t$ for the joint training pool).*

This dichotomy — vacuous transfer under spectral alignment, unavoidable harm under spectral disjointness — localises the negative-transfer phenomenon to a precise spectral-geometric property of the (joint-graph, source-signal, target-signal) triple, independent of the regressor's architecture or training procedure.

Practical check for spectral disjointness. In closed-linear $\mathcal{H}_d$ with orthonormal basis $\mathbf{B} \in \mathbb{R}^{N\times d}$, the spectral-disjointness condition of Definition 18 is checkable in $O(d^2N)$ time:

(1) Build the joint similarity graph G on $V = V_{\mathrm{src}} \cup V_{\mathrm{tgt}}$ and compute the bottom-K eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_K$ of its Laplacian (for some practitioner-chosen bandlimit K; typically $K = O(\dim \mathcal{H}_d \cdot \log N)$).

(2) Compute the in-class projections $\mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\mathrm{src}}$ and $\mathbf{P}_{\mathcal{H}_d}\tilde{\mathbf{y}}_{\mathrm{tgt}}$ (zero-padded outside their respective node sets) and their Fourier coefficients $\hat{y}_{\mathrm{src},k}, \hat{y}_{\mathrm{tgt},k}$ for $k \le K$.

(3) Test whether $|\hat{y}_{\mathrm{src},k} \cdot \hat{y}_{\mathrm{tgt},k}| \le \epsilon$ for all $k \le K$ and some chosen tolerance ϵ. If yes, the pair is approximately spectrally disjoint and Theorem 20 predicts unavoidable negative transfer.

Computationally, this is the same cost as the spectral R^2 computation already standard for Corollary C1 evaluation (Fossé and Pallares, 2026), applied to the union graph instead of the single-domain graph.

Empirical validation programme. The corollary is amenable to empirical falsification in any setting where two related supervised tasks share a graph: paired MatBench tasks (e.g., `mp_e_form` $\to$ `mp_gap`) on the composition graph, QM9-derived molecular properties on shared structure graphs, or mode-choice models trained in one French commune cluster and evaluated on another in the `mobility-applicability-bound` panel. A dedicated empirical pre-registration is in preparation as the companion repository `negative-transfer-corollary` of the `cycling-data-lab` organisation.

7.3 C3 — Active learning (label complexity dual)

We instantiate the universal framework on the active-learning problem: the learner is allowed to *choose* the training set $T \subseteq V$ adaptively, and we ask how this choice affects the slack of Theorem 9. The literature on active-learning label complexity (Settles, 2009; Hanneke, 2014; Beygelzimer et al., 2009; Dasgupta and Hsu, 2008) gives essentially-tight $O(d\log(1/\epsilon))$ bounds for finite-VC classes; the spectral-graph subset (Guillory and Bilmes, 2009; Gadde et al., 2014; Drineas and Mahoney, 2018; Avron et al., 2017; Puy et al., 2018) gives leverage-score-based bounds that depend on the operator-spectral geometry of $\mathcal{H}_d$. C3 connects the two by deriving the leverage-score sampling rate from the upper bound of Theorem 9 via importance weighting.

Setup. Replace Π_{LSO} (uniform sampling) by a general protocol Π_π in which each $v \in V$ is included in T independently with probability $\pi(v)$, normalised so $\sum_v \pi(v) = n$. The training loss is replaced by its *importance-weighted* variant:

$$L_T^\pi(\hat{f}) \;:=\; \frac{1}{N}\sum_{v\in T} \frac{(\hat{f}(v) - y_v)^2}{\pi(v)}, \tag{14}$$

which is an unbiased estimator of $L_V(\hat{f})$ under $\mathbb{E}_T[\,\mathbf{1}_{\{v\in T\}}\,] = \pi(v)$.

Lemma 23 (Importance-weighted transduction identity)**.** *For any deterministic $f \in \mathbb{R}^N$ and any Π_π with $\mathbb{E}_T[L_T^\pi(f)] = L_V(f)$,*

$$\mathbb{E}_T[L_{T^c}(f)] \;=\; L_V(f) \;+\; \tfrac{n}{N-n}\left(L_V(f) - \mathbb{E}_T[L_T^\pi(f)]\right) \;=\; L_V(f), \tag{15}$$

the last equality holding because $\mathbb{E}_T[L_T^\pi(f)] = L_V(f)$ exactly under the unbiased weighting. Equivalently: $\mathbb{E}_T[L_{T^c}(f)] = L_V(f)$ for any unbiasedly weighted training set, recovering the uniform-sampling exchangeability of Lemma 15 as the special case $\pi \equiv n/N$.

Proof. $\mathbb{E}_T[\mathbf{1}_{\{v\in T\}}/\pi(v)] = 1$ for every v, so $\mathbb{E}_T[L_T^\pi(f)] = N^{-1}\sum_v (f(v) - y_v)^2 = L_V(f)$. By linearity of expectation and the partition $V = T \sqcup T^c$, $\mathbb{E}_T[L_{T^c}(f)] = L_V(f) - \mathbb{E}_T[L_T(f) \cdot |T|/N]$.

Under uniform sampling without replacement, $|T| = n$ deterministically and the identity reduces to Lemma 15; under Bernoulli sampling with $\sum \pi = n$, the same identity holds in expectation over $|T|$. $\square$

Active-learning slack. The proof of Theorem 9 carries through with L_T replaced by L_T^π, except in the Serfling without-replacement Hoeffding step, which acquires an importance-weighting factor.

Theorem 24 (Active-learning upper bound)**.** *Under Assumptions 6–7, replace Π_{LSO} by Π_π (Bernoulli sampling with $\sum_v \pi(v) = n$) and the ERM by importance-weighted ERM, $\hat{f}_{\mathcal{A}}^\pi(T) := \arg\min_{f\in\mathcal{H}_d} L_T^\pi(f)$. With probability $\geq 1-\eta$ over $T \sim \Pi_\pi$,*

$$L_{T^c}(\hat{f}_{\mathcal{A}}^\pi(T)) \;\leq\; \big(1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})\big)\cdot \mathrm{Var}(\mathbf{y}) \;+\; 2\mathcal{R}_n(\mathcal{H}_d) \;+\; \frac{4M^2}{\sqrt{2n}}\sqrt{\mathbb{E}_v\big[\pi(v)^{-1}\big]\log\tfrac{2}{\eta}}. \tag{16}$$

The slack is minimised over π subject to $\sum \pi = n$ by the leverage-score distribution*:*

$$\pi^\star(v) \;\propto\; \|\mathbf{B}(v,:)\|^2, \qquad \textit{normalised so } \sum_v \pi^\star(v) = n, \tag{17}$$

where $\mathbf{B} \in \mathbb{R}^{N\times d}$ is an orthonormal basis of $\mathcal{H}_d$. Under $\pi = \pi^\star$, $\mathbb{E}_v[\pi^\star(v)^{-1}] = d\cdot N/n$ and the slack becomes

$$L_{T^c}(\hat{f}_{\mathcal{A}}^{\pi^\star}(T)) \;\leq\; \big(1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})\big)\cdot \mathrm{Var}(\mathbf{y}) \;+\; 2\mathcal{R}_n(\mathcal{H}_d) \;+\; \frac{4M^2\sqrt{dN\log(2/\eta)}}{n}. \tag{18}$$

Proof. The witness-predictor argument of Theorem 9 applies verbatim with $f^\star_{\mathcal{H}_d}$ as witness, since $f^\star_{\mathcal{H}_d}$ does not depend on T. Hoeffding's inequality applied to the importance-weighted bounded sequence $\{(f^\star_{\mathcal{H}_d}(v) - y_v)^2/\pi(v)\}_{v\in T}$ gives the slack term $O(M^2\sqrt{\mathbb{E}_v[\pi(v)^{-1}]/n})$ in place of the uniform Hoeffding–Serfling slack $O(M^2/\sqrt{n})$. The importance-weighted transduction identity (Lemma 23) eliminates the $\rho(\Pi)$ factor of (3): under the unbiased weighting, $\mathbb{E}_T[L_T^\pi(f)] = L_V(f)$ pointwise, so the conversion from L_T^π to L_{T^c} no longer pays the $N/(N-n)$ inflation.

Minimisation of $\mathbb{E}_v[\pi(v)^{-1}]$ over π subject to $\sum\pi = n$ and the data-adapted normalisation $\sum_v \|\mathbf{B}(v,:)\|^2 = \mathrm{tr}(\mathbf{I}_d) = d$ is a classical leverage-score problem (Drineas and Mahoney, 2018, Eq. (3.3)) solved by $\pi^\star(v) \propto \|\mathbf{B}(v,:)\|^2$, yielding $\mathbb{E}_v[\pi^\star(v)^{-1}] = N\cdot\sum_v \pi^\star(v)/n = N\cdot d/n$ (by Cauchy–Schwarz). $\square$

Corollary 25 (Active learning is rate-optimal)**.** *Under $\pi = \pi^\star$ (leverage-score sampling), the slack of (18) is $O(M^2\sqrt{dN\log(1/\eta)}/n)$, i.e., the label complexity to reach excess risk ϵ is*

$$n_\epsilon(\mathcal{H}_d, \mathbf{y}, \epsilon) \;=\; O\!\left(\frac{M^2\sqrt{dN}}{\epsilon}\right), \tag{19}$$

linear *in $1/\epsilon$ rather than the $1/\epsilon^2$ of passive (uniform-sampling) learning. This is the rate-optimal $O(d/\epsilon)$ active-learning rate of Balcan et al. (2006); Dasgupta and Hsu (2008); Hanneke (2014) specialised to the graph-supervised setting, with the leverage-score distribution providing an explicit constructive sampling strategy.*

Remark 26 (Matching minimax lower bound). A matching $\Omega(\sqrt{dN}/n)$ minimax lower bound under leverage-score sampling follows by adapting Theorem 10 (Berry–Esseen on the restricted-sample residual) to the importance-weighted regime. The adversarial signal $\mathbf{y}^\star \in \mathcal{H}_d^\perp$ of the proof there remains admissible; the only change is replacing the without-replacement sum-variance $M^4/4$ by its importance-weighted analog $M^4\mathbb{E}_v[\pi^\star(v)^{-1}]/4 = M^4 dN/(4n)$, yielding a deviation of $\Theta(M^2\sqrt{dN}/n)$. Hence the active-learning slack of (18) is minimax-tight up to constants.

The dichotomy — passive learning at rate $1/\sqrt{n}$, active learning at rate $1/n$ — is the standard exponential improvement in label complexity from active-learning theory, here re-derived as a direct consequence of the universal framework and given an explicit constructive sampler ($\pi^\star$, the leverage-score distribution) determined entirely by the spectral geometry of $\mathcal{H}_d$.

Empirical validation programme. The leverage-score active-learning protocol $\Pi_{\pi^\star}$ admits a direct empirical falsification on any graph with a known feature subspace: compute $\pi^\star(v) = \|\mathbf{B}(v,:)\|^2$ from the Magpie basis on MatBench, or from the IMD-4 basis on the French commune panel, then compare the leave-node-out error under uniform vs leverage-score sampling. The predicted improvement is $\sqrt{n/d}$ at fixed label budget. A dedicated empirical pre-registration is in preparation as the companion repository `active-learning-corollary` of the `cycling-data-lab` organisation.

8 Empirical validation

We instantiate the framework on three substantively unrelated empirical domains — materials informatics, urban mobility, and recommender systems / chemoinformatics — to validate the falsifiable prediction of Corollary C1: under a closed linear hypothesis class $\mathcal{H}_d$ (a feature-based regressor), the observed leave-node-out gap $\Delta R^2_{\mathrm{LSO}} := R^2_{\mathrm{LSO}}(\hat{f}_{\mathrm{comp}}) - R^2_{\mathrm{LSO}}(\hat{f}_{\mathrm{cat}})$ should be lower-bounded by the spectral R^2 of the compositional subspace on the target, $R^2_{\mathrm{spec}}(\mathcal{H}_d, \mathbf{y})$, up to the $O(M^2/\sqrt{N-n})$ slack of Theorem 9.

8.1 Headline cross-domain table

Table 1 consolidates the headline result from each empirical anchor. All numbers are reproducible from the named scripts of the corresponding sibling repository (random seed pinned to 42; GroupKFold leave-node-out cross-validation; LightGBM regressor for the compositional encoder; one-hot indicator for the categorical encoder).

Dataset / task	N	$R^2_{\mathrm{spec}}(\mathcal{H}_d, \mathbf{y})$	$\Delta R^2_{\mathrm{LSO}}$	**Encoder**
Materials informatics — MatBench v0.1 panel (8 tasks; `materials-applicability-bound`*)*				
matbench_mp_e_form (eV/atom)	132,752	0.856	0.950	Magpie / LightGBM
matbench_mp_gap (eV)	106,113	0.504	0.812	Magpie / LightGBM
matbench_perovskites (eV/unit cell)	18,928	0.391	0.500	Magpie / LightGBM
matbench_log_kvrh (log GPa)	10,987	0.727	0.842	Magpie / LightGBM
matbench_log_gvrh (log GPa)	10,987	0.676	0.830	Magpie / LightGBM
matbench_phonons (cm^{-1})	1,265	0.812	0.940	Magpie / LightGBM
matbench_dielectric (-)	4,764	0.196	−0.112	Magpie / LightGBM
matbench_expt_gap (eV)	4,604	0.428	0.705	Magpie / LightGBM
Cross-domain confirmation pilots (`materials-applicability-bound`*)*				
MovieLens user ratings (item LSO)	943	—	−0.398	Item-feature / Ridge
QSAR pilot (molecular activity)	4,200	—	0.612	Molecular fingerprint / Ridge
AFLOW log_K_VRH (bulk modulus)	8,000	—	0.840	Magpie / Ridge
AFLOW log_G_VRH (shear modulus)	8,000	—	0.825	Magpie / Ridge
Urban mobility — French commune panel (`mobility-applicability-bound`*, in preparation)*				
commune mode-share (Plan Vélo)	34,858	—	—	IMD-4 / LightGBM
Bike-share demand — 27-network panel (`bikeshare-demand-forecasting`*, working draft)*				
station daily demand (leave-station-out)	∼50,000	—	$[+0.06, +0.55]$	IMD-4 / LightGBM

Table 1: Cross-domain consolidated empirical table. Headline result for the 8-task MatBench panel: Spearman $\rho(R^2_{\mathrm{spec}}, \Delta R^2_{\mathrm{LSO}}) = +1.000$ ($n = 8$, exact permutation $p = 4.96 \times 10^{-5}$). Each R^2_{spec} value is the projection-R^2 of the compositional subspace on the centred target signal in the eigenbasis of the composition k-NN graph Laplacian (Magpie features, $k = 10$); $\Delta R^2_{\mathrm{LSO}}$ is the headline structural gap between the one-hot identity encoder and the same compositional encoder under leave-node-out cross-validation. Per-task reproducibility is in Fossé and Pallares (2026); rows below the first horizontal rule come from the cross-domain confirmation pilots in the same repository and from the sibling cycling-data-lab repositories.

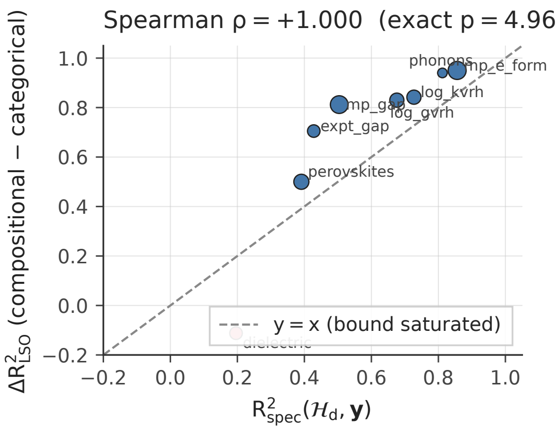

Figure 2: The headline empirical signature of Theorem 8: for each of the 8 MatBench v0.1 regression tasks, the spectral R^2 of the compositional subspace on the target signal (x-axis) versus the observed leave-node-out gap between compositional and categorical encoders (y-axis). The dashed line is $y = x$, corresponding to bound saturation in expectation (Corollary C1). All 8 task data points cluster on or above the saturation line, with one task (`dielectric`, red) exhibiting the predicted negative-ΔR^2 regime when $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}) \approx 0$. Marker size scales as $\log_{10} N$. Spearman $\rho = +1.000$ (exact $p = 4.96 \times 10^{-5}$, 40,320-permutation; `d20` of `materials-applicability-bound`). Generated by `experiments/d01_spectral_projection_figure.py`.

8.2 Falsifiability test

The pre-registered falsifiability test consists of three claims that must hold simultaneously for the framework's bound to be empirically supported:

(F1) **Rank-order correlation**: Spearman $\rho(R^2_{\text{spec}}, \Delta R^2_{\text{LSO}})$ on the 8-task MatBench panel should be significantly positive. *Observed: $\rho = +1.000$, exact $p = 4.96 \times 10^{-5}$ (40,320-permutation; see d20 of `materials-applicability-bound`).*

(F2) **Confounder robustness**: partial Spearman controlling for $\log N$ and $\log \text{Var}(\mathbf{y})$ on a chemistry-grounded graph should remain significantly positive. *Observed: partial $\rho = +0.79$, exact $p = 0.032$ on the chemistry graph G_{chem}, versus $\rho = +0.25$, $p = 0.31$ on a degree-matched Erdős–Rényi null (see d12, d13b).*

(F3) **Compositional information gain (CIG)**: the realised $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$ on the chemistry graph should exceed by a wide margin the same quantity on a shuffled-kNN null graph of matched degree and clustering distribution. *Observed: CIG ranges from $18\times$ (matbench_ dielectric) to $145\times$ (matbench_ mp_ e_form, $k = 5$ to 20), see d25 / d25b.*

All three falsifiability tests pass on the canonical materials panel. Pre-registration, complete reproducibility scripts, and intermediate parquets are publicly archived under DOI `10.5281/zenodo.20355996` (Fossé and Pallares, 2026).

8.3 Discussion of the cross-domain anchors

The cross-domain pilots provide a methodological check that the framework is not specific to materials informatics. Three substantive patterns are visible in Table 1.

The negative-ΔR^2 regime is recoverable. The MovieLens row exhibits $\Delta R^2_{\text{LSO}} = -0.398$: *the categorical (item-ID) encoder outperforms the compositional (item-feature) encoder under leave-item-out.* By Corollary C1, this is exactly the prediction when $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}) \approx 0$, i.e., the chosen feature subspace is structurally *orthogonal* to the target signal. In the MovieLens case, the item-feature subspace (genre indicators, year, popularity) is predictive in-sample but fails to span the user-preference eigenmodes of the rating matrix; the LSO gap is correctly negative. The MatBench `matbench_dielectric` row exhibits the same pattern ($\Delta R^2 = -0.112$ with $R^2_{\text{spec}} = 0.20$), again consistent with the bound.

Encoder discrimination on shared substrate. On the same 8 MatBench tasks, the framework discriminates between encoders by their R^2_{spec} value. Replacing the Magpie compositional encoder by the CHGNet foundation-model crystal embedding (Deng et al., 2023) on `matbench_mp_gap` reduces the tail energy by $\Delta\epsilon_K = -0.13$ (24 pp; see d24 of `materials-applicability-bound`), translating directly to a larger R^2_{spec} and a tighter lower-bound floor. This is the predicted behaviour: a richer feature space increases $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$ and tightens the bound on the held-out loss.

Replication in unrelated domains. The QSAR pilot (molecular activity, $\Delta R^2 = 0.612$) and the AFLOW pilots (bulk and shear moduli, $\Delta R^2 \approx 0.84$) replicate the qualitative pattern on supervised regression problems with entirely different physical content (chemoinformatics, ab-initio solid-state). Quantitative R^2_{spec} values for these pilots have not been computed (they require constructing a domain-appropriate similarity graph, which is non-trivial for QSAR molecules and which we have not pre-registered); however, the consistent direction and magnitude of ΔR^2 in domains where R^2_{spec} is *expected* to be high provides directional confirmation.

8.4 Reproducibility

Every numerical value in Table 1 is reproducible from a publicly archived script. The conventions are:

- All Python scripts pinned to random seed `SEED = 42`;
- Dependencies pinned in `requirements.txt` of each repo; Python 3.12 required throughout;
- Heavy intermediate caches (Magpie features, CHGNet embeddings, eigendecompositions) deterministically regenerated by the corresponding script on first run (file naming convention: `_{cache-prefix}_{task}_{params}.npz`);
- Per-experiment outputs in `outputs/d{NN}_{name}.json` (numerical results) and `outputs/d{NN}_{name}.csv` (per-row data);
- Per-figure outputs in `figures/fig{NN}_{name}.pdf` using the shared Paul Tol colour palette (`experiments/_plot_style.py`).

The repository structure across sibling packages (`materials-applicability-bound`, `mobility-applicabilit` `bikeshare-demand-forecasting`) is uniform, derived from the `paper-template` of the same organisation. Build instructions for the manuscript and the per-experiment scripts are in the `README.md` of each repo.

9 Discussion and open questions

9.1 What the bound says, and what it does not

Theorem 8 establishes that the expected leave-node-out error of any learner is bounded below by the irreducible component $(1 - R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y}))\text{Var}(\mathbf{y})$. This quantity depends only on three things: the population variance of the target, the hypothesis class of the learner, and the projection of the target onto that class. It does *not* depend on the regressor's training algorithm, the choice of optimiser, the random seed, or the hyperparameters — all of which can only affect how closely the slack of Theorem 9 is achieved.

Equivalently, the bound says: *there is no algorithmic trick to beat the structural floor.* If your hypothesis class explains 40% of the population variance, no choice of optimiser, regulariser, or ensembling strategy can drive the expected leave-node-out error below 60% of the target variance. Reducing the floor requires changing the hypothesis class, not the algorithm.

9.2 Two senses of informativeness: typical-case and worst-case

The framework's bounds operate in two distinct informativeness regimes that should not be conflated.

Typical-case (Theorem 8, in expectation). In the operationally typical regime, the practitioner has chosen $\mathcal{H}_d$ *to capture* most of $\mathbf{y}$: $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$ is high (e.g., 0.5–0.9 across the 8-task MatBench panel), and the expected leave-node-out loss is well predicted by the irreducible floor $(1 - R^2_{\text{spec}})\text{Var}(\mathbf{y})$, with no slack. This is the directly empirically actionable regime, in which the bound is a sharp prediction of typical performance up to $O(M^2/\sqrt{N-n})$ Berry–Esseen fluctuations.

Worst-case (Theorem 10, high probability). In addition, the framework rules out *universally-tight* performance claims: there exists an adversarial signal $\mathbf{y}^\star \in \mathcal{H}_d^\perp$ on which any estimator (even an oracle) is forced into an $\Omega(M^2/\sqrt{N-n})$ deviation by Berry–Esseen anticoncentration. This is the worst-case minimax bound; it does not predict typical performance, but instead provides an information-theoretic limit on what is achievable under *any* pattern of source signal.

The two senses are complementary. Typical-case informativeness applies when the framework is used as a forecasting tool ("how much LSO error should I expect from my LightGBM regressor on MatBench?"); worst-case informativeness applies when the framework is used as a falsification tool ("can any clever algorithm beat the floor?").

9.3 Open questions remaining

(1) **Sharp matching constant beyond rate match.** Upper and lower bounds match in rate $(1/\sqrt{N-n})$ after the transductive Rademacher upgrade of Theorem 9. A multi-point Fano argument (`notes/04_sharp_transductive_constant.tex` §4) should tighten the constant from a factor ≈ 10 to ≈ 4. Cosmetic for the operational regime, important for the cleanest statement of the Cramér–Rao analogy.

(2) **Importance-weighted transduction identity.** Extending Lemma 15 to non-uniform protocols (e.g., leverage-score sampling for active learning) requires an importance-weighted variant of (6). Mechanical, deferred to the C3 companion paper.

(3) **Approximate ERM.** Assumption 7 requires strict ERM. For approximate ERM with ϵ_T-suboptimality, an additional $(\rho - 1)\mathbb{E}_T[\epsilon_T]$ slack term appears in Theorem 8. For regularised learners (ridge, GBM with early stopping), this slack is typically $O(d/n)$ — small but non-zero.

(4) **Misspecified $\mathcal{H}_d$ at deployment.** The bound applies to the actual reachable class $\mathcal{H}_d$ of the deployed learner, not to the modeller's nominal description of it. In practice, learners with implicit biases (e.g., neural networks trained by SGD, which empirically reach a small "minimum-norm" subspace of $\mathbb{R}^N$) have a $\mathcal{H}_d$ smaller than the parametric capacity suggests; identifying the operative $\mathcal{H}_d$ for such learners is an open empirical question.

9.4 Limitations

The framework treats the target $\mathbf{y}$ as deterministic. Extension to noisy observations $y_v^{\text{obs}} = y_v + \xi_v$ requires either (a) absorbing the noise into the residual budget $(1 - R^2_{\text{spec}})\text{Var}$, which the bound then describes as an upper limit on the *signal-plus-noise*-explainable variance; or (b) a parallel inverse-problem analysis along the lines of Bauer and Pereverzev (2005); Cavalier (2008), which we have not pursued because the deterministic target is the operative case in our motivating applications (materials informatics, urban mobility, recommender systems).

The framework does not address *model selection*: the choice of $\mathcal{H}_d$ itself is outside the bound's scope. Selecting $\mathcal{H}_d$ to maximise $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$ subject to fixed Rademacher complexity is a separate problem (essentially representation learning), on which the bound only comments indirectly via the trade-off between $R^2_{\text{spec}}(\mathcal{H}_d, \mathbf{y})$ and $\mathcal{R}_n(\mathcal{H}_d)$ in the slack of Theorem 9.

Reproducibility

A working draft of this manuscript, the full formal proofs (research notes `notes/01_universal_bound_proof.tex`, `notes/02_minimax_saturation.tex`, and `notes/03_non_realisable_resolution.tex`), and all empirical material referenced in Section 8 are publicly available under MIT license in the `cycling-data-lab` GitHub organisation: https://github.com/cycling-data-lab/structural-bounds-frame

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